Constraints Without Commitment: On the Limits of Algebraic Unification in Contemporary Deep Learning

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1 Introduction

Recent work in Categorical Deep Learning (CDL) proposes a unifying mathematical language for neural network design, capable of bridging the long-standing divide between top-down constraints and bottom-up implementation. By appealing to category theory—specifically to universal algebra in a 2-category of parametric maps—this framework offers a principled account of architectural invariance, recursion, and weight tying.

The ambition of this project is substantial and, in many respects, successful. Practices that previously appeared as engineering heuristics are reinterpreted as algebraic necessities. Entire classes of models—geometric, recurrent, and message-passing—are shown to instantiate a shared compositional structure.

This essay does not contest these achievements. Instead, it argues that the very success of categorical unification reveals a boundary. While CDL excels at formalizing *constraints* on computation, it does not formalize *commitment* in the sense required for agency, history, and accountability. The distinction is not cosmetic. It marks the limit of what can be expressed in purely algebraic terms.

The argument proceeds by strengthening the technical foundation of this claim, deepening its philosophical implications, and engaging constructively with the framework's genuine contributions.

2 Accountable Lawfulness

The preceding analysis suggests a precise characterization of the systems described by categorical deep learning. They are maximally lawful and minimally accountable. They perfectly respect algebraic priors, internalize structure, and generalize efficiently. At the same time, they lack the internal resources to record refusal, bear loss, or bind themselves to history.

Non-invertible computation allows such systems to destroy information, but not to incur obligation. Actions change representations, but they do not settle futures. The result is a form of intelligence that participates in computation but not in history.

This diagnosis does not undermine the value of categorical unification. Rather, it clarifies its scope. The framework explains how systems act. It does not explain why they never give, refuse, or stop. That gap is not accidental; it is the mark of a boundary between lawful structure and accountable agency.

3 What Categorical Deep Learning Gets Right

Before turning to critique, it is essential to acknowledge the genuine advances offered by the categorical approach.

First, CDL achieves a real unification. Prior to this work, geometric deep learning, recursive architectures, and weight sharing were treated as distinct design paradigms. CDL demonstrates that these are instances of a common algebraic structure: structure-preserving maps between parameterized computational spaces. This clarification is not merely aesthetic; it explains why certain designs generalize well and others fail.

Second, the framework explains the remarkable data efficiency of models with built-in invariances. By encoding permutation equivariance or other symmetries directly into the architecture, the hypothesis space is dramatically reduced. Empirically, this yields near-exponential reductions in sample complexity—a result that is both theoretically satisfying and practically significant.

Third, the identification of the *carry problem* in graph neural networks represents a nontrivial technical insight. The difficulty GNNs have with discrete, cascading transitions is not obvious from standard formulations. The connection to nontrivial fibrations in topology underscores the mathematical depth of the observation.

These achievements justify taking the framework seriously on its own terms.

4 Algebraic Constraints and Their Scope

To understand the limits of categorical unification, it is necessary to examine the progression of algebraic structures it employs.

4.1 From Groups to Categories

Definition 1. *A* group *is a set G equipped with a binary operation $\cdot$ satisfying closure, associativity, the existence of an identity element, and invertibility.*

Group actions underlie much of geometric deep learning. Rotations, reflections, and permutations act on inputs in ways that can always be undone. No information is lost.

Definition 2. *A* monoid *is a set M with an associative binary operation and an identity, but without the requirement of invertibility.*

Monoids naturally model computation. Algorithms such as shortest-path solvers map many distinct inputs to a single output, destroying information irreversibly.

Definition 3. *A* category *consists of objects, morphisms between objects, identity morphisms, and associative composition, without requiring that all morphisms be composable or invertible.*

Categories generalize both structures, allowing heterogeneous composition and non-invertible processes.

4.2 Quotients and Erasure

Consider Dijkstra's algorithm. Distinct weighted graphs $G_1, G_2, \ldots$ may all map to the same shortest-path tree. Formally, the algorithm defines a quotient map from the space of graphs to equivalence classes determined by path length. The categorical formalism captures this quotienting cleanly.

What it does not capture is which representative was chosen. The computation is lawful, but its result erases history. This distinction—between preserving structure and preserving commitment—will recur.

4.3 Lawfulness vs. Commitment

Structure-preserving maps define *lawful computation.* Commitment, by contrast, involves selection from equivalence classes with consequences for future action. Algebraic invariance can describe the former, but it is silent on the latter.

5 Constraints as Homomorphism Requirements

Within the categorical framework, a constraint is not a numerical restriction on parameters but a semantic requirement: a neural layer must act as a homomorphism between algebras. This reframes architectural design as the problem of selecting endo-functors whose algebras encode the intended computational behavior.

In this sense, constraints remain meaningful even under reparameterization. Two-morphisms in the 2-category of parametric maps permit parameters to vary, but only insofar as the induced morphism preserves the algebraic structure. A reparameterization that violates the homomorphism condition is not merely suboptimal; it is ill-typed within the theory.

This interpretation clarifies why categorical deep learning distinguishes between syntactic freedom and semantic rigidity. Syntax may vary freely, but semantics—captured by the algebra—must remain invariant. The framework therefore addresses the question of how computation is shaped, not how future action is bound.

The distinction is decisive. Homomorphism requirements constrain what a system *can compute*, but they do not constrain what a system *must stand behind.* Constraint, in this sense, is alethic rather than deontic: it governs possibility, not obligation.

6 Recursion, Folding, and the Erasure of History

Recursion is modeled categorically as a fold: a homomorphism from an inductively defined data structure to a result. In this view, the history of a computation is preserved only insofar as it is encoded in the structure of the input. The execution trace itself is erased in favor of the final value.

Similarly, the unrolling of recurrent networks via copy maps encodes repetition rather than lineage. Weight tying ensures that the same computation is applied at each step, but it does not record the fact that a particular path was taken or refused. The endofunctor governing the recurrence specifies lawful evolution, not historical memory.

What survives is structure; what disappears is biography.

7 Internalized Reasoning and the Absence of Obligation

A central claim of the categorical approach is that models should internalize algorithms rather than rely on external tools. Internalization yields stability, efficiency, and correctness guarantees. Carrying in addition or conservation laws in physics should be reflected in the model's internal geometry, not recovered through repeated querying.

However, what is internalized are procedures, not consequences. Cost appears as a computational budget, not as an obligation. Correctness is mathematical, not normative. The framework provides no mechanism by which a system could absorb risk, incur debt, or be worse off as a result of its actions.

Internalized reasoning remains non-dissipative.

8 Internalization, Stability, and the Limits of Consequence

The categorical program emphasizes the internalization of algorithms as a source of stability and efficiency. Internalized reasoning avoids the brittleness of external tool calls, which require accurate input prediction and repeated re-invocation. In this sense, internalization reduces computational cost and improves robustness.

However, what is internalized is the procedure, not the consequence. Stability is achieved by aligning neural machinery with symbolic computation, not by encoding responsibility for failure. Correctness guarantees remain mathematical, expressed as post-conditions or invariants, rather than normative commitments.

The framework acknowledges uncertainty by suggesting that systems might be trained to back off when computational demands exceed their capacity. Yet such backing off remains instrumental. It is triggered by resource limits, not by a standing obligation to refuse.

Internalization thus stabilizes reasoning without internalizing stake. The system becomes better at acting correctly, not at being bound by having acted.

9 The Carry Problem and Emergent Obligation

The carry operation in arithmetic provides a concrete illustration of the distinction between local structure and global consequence.

Adding digits modulo 10 treats each digit independently. Ordinary addition, however, is not independent: a transition from $9 \to 0$ in one digit forces an increment in the next. The obligation is not in the state, but in the transition.

Graph neural networks struggle with such phenomena because message passing captures local state changes but not cascading obligations. The Hopf fibration $S^3 \to S^2$, with fiber S^1, provides a continuous analogue: the total space is not decomposable into independent components.

Once a carry occurs, it cannot be undone without altering the entire structure. The categorical framework captures the algebraic structure of carrying, but it does not record the fact that a carry has occurred as a binding historical event.

10 Parametric Maps and Asymmetric Loss

The 2-category of parametric maps, Para, makes parameters explicit objects of study. Objects are pairs (A, P), where A is a space and P a parameter space. Morphisms are parameterized functions $f : P \times A \to B$. Two-morphisms correspond to reparameterizations preserving structure.

Weight tying arises naturally via copy maps. This symmetry explains why shared parameters induce coordinated behavior.

However, this symmetry also reveals a limitation. There is no notion of ownership or unilateral commitment. All admissible reparameterizations that preserve structure are equally valid. Loss functions update parameters symmetrically.

In game theory, commitment involves irrevocably constraining one's own future actions to change others' incentives. This requires burned bridges: regions of parameter space that cannot be re-entered. Para contains no such mechanism.

Loss exists as information destruction, but no agent bears it.

11 Symmetry of Parameters and the Impossibility of Ownership

The 2-category of parametric maps makes parameters explicit but treats them as neutral mathematical objects. Morphisms relate parameterized spaces, and two-morphisms reparameterize them while preserving structure. At no point does the framework assign control, ownership, or agency to a particular party.

Asymmetry arises only at the level of computation: non-invertible morphisms destroy information, and certain natural transformations exist in one direction but not the other. This asymmetry, however, is internal to the diagram. It does not imply that any agent bears unilateral loss.

Loss is therefore endogenous to the morphism rather than exogenous to an agent. Data may be discarded, but no one is worse off. Unilateral commitment—understood as the irrevocable restriction of one's own future actions—is not representable within this symmetric parameter space.

This explains why cooperation in the framework is purely compositional. Joint behavior is correct when it is isomorphic to the composition of parts. What is missing is the possibility of binding oneself so that others may rely on that binding.

12 Refusal as a Missing Primitive

Optimization frameworks permit weak negation: actions are avoided because they are costly. Such negation can always be overridden with sufficient resources.

Refusal, by contrast, is strong negation: an action is forbidden regardless of utility. This distinction maps onto the philosophical divide between alethic modality (what is possible) and deontic modality (what is permitted).

Category theory is fundamentally alethic. It describes what transformations are lawful. Commitment and refusal are deontic; they persist independently of possibility.

Refusal also requires history. Having refused an action at time t_1 must constrain options at t_2. Absent a state that records refusal, the system remains perpetually revisable.

Purely structural models thus inhabit a form of permanent bad faith: past choices never bind the present.

13 Weak and Strong Negation

The framework permits a form of weak negation. A system may decline to answer a question because it exceeds a computational budget or violates a learned prior. Such negation is contingent and revisable. With sufficient resources or a change in context, the refusal can be overridden.

Strong negation is categorically different. To refuse in the strong sense is to forbid an action regardless of cost. Such refusal persists across contexts and binds future behavior. It is deontic rather than alethic.

Categorical constraints are necessarily weak in this sense. They describe what cannot occur given the structure of the computation, not what must not occur despite being possible. Treating refusal as first-class would therefore require a state that records negation as an event, not merely as the absence of a morphism.

Introducing such states would break the assumption that computation always composes. Some morphisms would lead to terminal conditions that are not identities but prohibitions.

14 Time, Reversibility, and Worldhood

Categories naturally model processes: continuous transformations composed over time. Events, however, are breaks—partitions of time into before and after.

Optimization lacks terminality. Local minima are not ends; they are pauses. Games, by contrast, have terminal states where no further moves are legal.

Non-invertible morphisms capture logical irreversibility, but not temporal irreversibility. Two identical states reached via different histories are structurally indistinguishable, yet historically distinct.

Worldhood requires stake. A world is not merely a state space, but a space of concern in which actions matter. Structure without stake yields process without world.

15 Cooperation, Commitment, and Common Knowledge

Shared parameters enable structural coordination, but not trust. If agents can secretly reparameterize, cooperation collapses.

Game-theoretic commitment requires common knowledge of irreversible moves. Agreements without enforcement are, as Hobbes observed, merely words.

Categorical alignment provides the grammar of cooperation, but not its binding. This explains why external tools and verifiers remain necessary: the system cannot commit to itself.

16 Accountable Lawfulness

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17 Analysis

Categorical Deep Learning succeeds at unifying constraints and implementation. It formalizes intelligence as lawful structure and explains why certain architectures generalize so effectively.

What it does not formalize is commitment. Constraints shape computation, but they do not bind future action. Irreversibility appears as information loss, not obligation. Refusal remains instrumental rather than principled.

If computation explains how systems act, it does not explain why they never give. Crossing that boundary requires a theory in which actions can settle, costs can be borne, and history can accumulate.

18 From Lawful Computation to Eventful Structure

The preceding analysis has shown that Categorical Deep Learning provides a complete account of lawful computation: it unifies constraints and implementations by identifying architectures with structure-preserving maps. Within this framework, invariance, recursion, weight tying, and non-invertible computation are explained as algebraic necessities rather than engineering choices.

At the same time, this completeness marks a boundary. The categorical formalism is intentionally indifferent to how a computation is realized in time, to which events occurred, and to whether alternatives were ever explicitly refused. Reparameterization preserves lawfulness precisely because the framework is designed to quotient away implementation history.

What is required to move beyond this boundary is not a revision of the algebraic account, but a change in level. Rather than enriching architectures directly, one can descend to a description in which implementations are treated as eventful processes, and only subsequently recover algebraic structure by abstraction. In such a setting, histories are primary rather than incidental, and irreversibility appears as an explicit feature of state rather than as information loss alone.

This motivates the introduction of a field-theoretic intermediate description in which scalar accumulations, vector flows, and entropy production summarize the thermodynamic consequences of event histories. From this level, both geometric constraints and categorical laws can be recovered as distinct abstractions that forget different aspects of the underlying dynamics.

The following sections make this claim precise. They show how an event-sourced implementation calculus gives rise to an RSVP-style field theory, and how Geometric Deep Learning and Categorical Deep Learning emerge as orthogonal quotients of that field-level description. In this way, accountability and commitment are not appended to categorical structure, but shown to arise from what categorical abstraction necessarily leaves behind.

19 From Eventful Implementations to Algebraic Architectures

The categorical deep learning framework is often presented as a bridge between constraints and implementation. In this work, however, the direction of derivation is reversed. Rather than treating categorical structure as foundational and implementations as instances, we begin from an event-sourced implementation calculus and show how both Geometric Deep Learning and Categorical Deep Learning arise as abstractions of a more primitive field-theoretic description.

At the base level lies an operational calculus in which structures are generated by irreversible events. Objects are not primitive sets or vector spaces but stabilized outcomes of event histories. Composition is temporal rather than algebraic, and identity corresponds to the absence of events rather than a neutral morphism. This level describes implementation in the strongest sense: what exists is only what has been constructed.

From this eventful substrate, a field-theoretic description emerges by coarse-graining over fine-grained event order. Distinct histories are identified when they induce the same scalar accumulations, vector flows, and entropy production. The result is a relativistic scalar–vector–entropy field theory in which dynamics are governed by conservation, flow, and dissipation rather than by discrete computational steps. At this level, commitment appears as irreversibility in the entropy field, and history is preserved only through its thermodynamic consequences.

Geometric Deep Learning is obtained by a further abstraction that forgets thermodynamic content while retaining symmetry structure. When RSVP field configurations are quotiented by group actions—permutations, rotations, or more general equivariances—the remaining structure specifies which transformations leave the dynamics invariant. Message passing, equivariant layers, and symmetry- respecting architectures arise as implementations that are compatible with this symmetry-reduced description. In this sense, geometric constraints are not primitive assumptions but residual structure left after discarding energetic and entropic detail.

Categorical Deep Learning arises by a different abstraction of the same RSVP substrate. Instead of quotienting by symmetry, one quotients by lawful replayability of computation. Event histories are identified when they implement the same abstract computation, regardless of parameterization or concrete realization. What survives this identification are algebras over endofunctors, parametric maps, and structure-preserving transformations between them. Weight tying, recursion, and non-invertible computation appear here as algebraic properties of stable implementations, not as foundational axioms.

On this view, Geometric Deep Learning and Categorical Deep Learning are not competing foundations but orthogonal projections of a common underlying theory. Both are derived from RSVP by forgetting different aspects of the eventful substrate: geometry forgets thermodynamic commitment, while category theory forgets concrete event history. Neither abstraction recovers the full structure of the underlying implementation calculus, but each captures a distinct class of invariants relevant to learning.

The ordering is therefore essential. RSVP is not recovered from categorical deep learning; categorical deep learning is recovered from RSVP by abstraction. What categorical deep learning unifies—constraints and implementation—has already been unified at the level of eventful dynamics. The categorical framework formalizes the lawful remainder after irreversible, field-level commitments have been erased.

20 Mathematical Derivation as a Chain of Quotients and Forgetful Functors

This section makes precise the claim that (i) a field-theoretic description arises as a coarse-graining of eventful implementations, and that (ii) both Geometric Deep Learning and Categorical Deep Learning arise as further abstractions of that field-level description. The construction is expressed as a sequence of functors that forget different aspects of structure.

20.1 Eventful implementations as a history category

Let Ev be a fixed set of primitive event types (e.g. POP, MERGE, LINK, COLLAPSE) equipped with admissibility conditions that determine which events may follow which.

Definition 4 (Histories and implementations)**.** *A* history *is a finite (or countable) admissible* *sequence* $h = (e_1, \dots, e_T)$ *with* $e_t \in \mathsf{Ev}$*. An* implementation state *is an object* x *produced by replaying a history from a distinguished initial state* x_0*, written* $x = \mathrm{Replay}(h; x_0)$*.*

We package histories into a category in the usual "free category" manner.

Definition 5 (History category Hist)**.** *Define a category* Hist *whose objects are implementation states and whose morphisms are histories that transform one state to another:*

$$h : x \to y \quad \textit{iff} \quad y = \mathrm{Replay}(h; x).$$

Composition is concatenation of histories (when admissible), and identities are empty histories.

Intuitively, Hist is the most concrete level: it remembers *which events occurred and in what order.* All later structures are obtained by forgetting some of this detail.

20.2 Coarse-graining to RSVP: thermodynamic equivalence

Let Fld denote a category of RSVP field configurations. For present purposes, an object of Fld may be taken as a triple

$$X = (\Phi, \mathbf{v}, S)$$

where Φ is a scalar field, $\mathbf{v}$ a vector field, and S an entropy (or irreversibility) field; morphisms in Fld are admissible field updates (e.g. time-evolution steps) compatible with the chosen dynamics.

Definition 6 (Thermodynamic coarse-graining functor)**.** *A* thermodynamic coarse-graining *is a functor*

$$\mathcal{Q}_{\mathrm{RSVP}} : \mathsf{Hist} \to \mathsf{Fld}$$

such that, for any state x*, the object* $\mathcal{Q}_{\mathrm{RSVP}}(x)$ *is the field configuration* $(\Phi_x, \mathbf{v}_x, S_x)$ *obtained by aggregating the history-sourced quantities of* x*, and for any history* $h : x \to y$*,* $\mathcal{Q}_{\mathrm{RSVP}}(h)$ *is the induced field update.*

The key property is that $\mathcal{Q}_{\mathrm{RSVP}}$ identifies many distinct histories as the same field trajectory.

Definition 7 (RSVP equivalence of histories)**.** *Define* $x \sim_{\mathrm{RSVP}} y$ *iff* $\mathcal{Q}_{\mathrm{RSVP}}(x) = \mathcal{Q}_{\mathrm{RSVP}}(y)$ *in* Fld*. Likewise define* $h \sim_{\mathrm{RSVP}} h'$ *iff* $\mathcal{Q}_{\mathrm{RSVP}}(h) = \mathcal{Q}_{\mathrm{RSVP}}(h')$*.*

Proposition 1 (Field-level description is a quotient of implementations)**.** $\mathcal{Q}_{\mathrm{RSVP}}$ *factors through the quotient of* Hist *by* $\sim_{\mathrm{RSVP}}$*. In particular, the field description forgets fine-grained event order and retains only its thermodynamic consequences.*

Proof. By definition, objects and morphisms identified by $\sim_{\mathrm{RSVP}}$ have the same image in Fld. Therefore $\mathcal{Q}_{\mathrm{RSVP}}$ is constant on equivalence classes and factors through the corresponding quotient construction. □

This is the formal sense in which RSVP is *derived* from implementation histories: it is what remains invariant under thermodynamic coarse-graining.

20.3 Deriving Geometric Deep Learning from RSVP by symmetry reduction

Let G be a group acting on the relevant field configurations (e.g. permutations of nodes, rotations, translations), written as

$$\rho : G \to \mathrm{Aut}_{\mathsf{Fld}}(X), \qquad g \mapsto \rho(g).$$

Definition 8 (Symmetry reduction / orbit functor)**.** *Define a functor*

$$\mathcal{Q}_{\mathrm{GDL}} : \mathsf{Fld} \to \mathsf{Fld}/G$$

sending a configuration X to its orbit $[X] = \{\rho(g)X : g \in G\}$ and sending a morphism $f : X \to Y$ to its induced map $[f] : [X] \to [Y]$ whenever this is well-defined (i.e. f is G-equivariant).

Architectures in Geometric Deep Learning may be characterized as maps that descend to the quotient and therefore respect these symmetries.

Definition 9 (Equivariant maps as admissible implementations)**.** *A map $f : X \to Y$ is G*-equivariant *if*

$$f(\rho(g)X) = \rho(g)f(X) \quad \textit{for all } g \in G.$$

Equivariant maps are precisely those that induce well-defined morphisms $[f] : [X] \to [Y]$ in Fld/G.

Proposition 2 (GDL as a quotient of RSVP)**.** *The symmetry-reduced description Fld/G forgets thermodynamic and entropic detail beyond what is required for equivariance, retaining only the invariance constraints. Equivariant neural layers are implementations of morphisms in Fld/G.*

Proof. Passing from X to $[X]$ identifies all G-related configurations; thus only invariants under G remain. A layer implements a morphism on orbits precisely when it is equivariant, i.e. it respects the quotient. □

Thus Geometric Deep Learning is obtained from RSVP by forgetting more: it discards field "substance" and retains symmetry-encoded constraints.

20.4 Deriving Categorical Deep Learning from RSVP by lawfulness quotients

To extract Categorical Deep Learning, we do *not* quotient by symmetry but by *computational lawfulness*. Informally, two implementations are identified when they realize the same abstract computation, regardless of parameterization or microscopic event history.

Let Comp be a category of abstract computations (types/objects and programs/morphisms) adequate to represent the target algorithmic structures (folds, automata, dynamic programs, etc.).

Definition 10 (Lawfulness quotient functor)**.** *A* lawfulness quotient *is a functor*

$$\mathcal{Q}_{\mathrm{CDL}} : \mathsf{Fld} \to \mathsf{Comp}$$

that maps a field configuration X to an abstract computational object $\mathcal{Q}_{\mathrm{CDL}}(X)$ and maps a field update to its induced computation, identifying any two realizations that satisfy the same pre/post-condition behavior (or equivalently, the same algebraic laws).

At this stage, parameters become explicit, and reparameterizations become 2-morphisms. One convenient formalization is via the 2-category of parametric maps.

Definition 11 (Parametric maps (sketch))**.** *Let* **Para** *have objects* (A, P)*, where* A *is an input space and* P *a parameter space. A 1-morphism* $(A, P) \to (B, Q)$ *is a parameterized map* $f : P \times A \to B$ *(with an appropriate parameter transport to* Q*), and a 2-morphism is a reparameterization* $\varphi : P \to P'$ *such that the induced maps agree up to* φ*.*

Within CDL, constraints are expressed as homomorphism conditions for algebras of endo-functors.

Definition 12 (Endofunctor algebras and constraint as homomorphism)**.** *Let* F *be an endofunctor on a category* $\mathcal{C}$*. An* F*-algebra is a pair* (A, α) *with* $\alpha : FA \to A$*. A morphism of* F*-algebras* $f : (A, \alpha) \to (B, \beta)$ *satisfies*

$$f \circ \alpha = \beta \circ Ff.$$

In CDL, an architectural constraint *is precisely such a homomorphism requirement.*

Proposition 3 (CDL as a quotient of RSVP distinct from GDL)**.** *The functor* $\mathcal{Q}_{\mathrm{CDL}}$ *identifies RSVP realizations by computational lawfulness (algebraic semantics), not by symmetry orbits. Hence* $\mathcal{Q}_{\mathrm{CDL}}$ *and* $\mathcal{Q}_{\mathrm{GDL}}$ *are, in general, incomparable: they forget different structure.*

Proof. $\mathcal{Q}_{\mathrm{GDL}}$ quotients by group actions, retaining invariants under G. $\mathcal{Q}_{\mathrm{CDL}}$ quotients by satisfaction of algebraic laws (e.g. fold laws, automaton laws), retaining invariants under reparameterization and implementation details. Since group invariants need not determine algebraic semantics and vice versa, neither quotient refines the other in general. □

20.5 Summary as a commuting extraction pattern

The derivation can be summarized as the existence of a coarse-graining from implementations to fields, followed by two distinct abstractions:

$$\mathsf{Hist} \xrightarrow{\mathcal{Q}_{\mathrm{RSVP}}} \mathsf{Fld} \xrightarrow{\mathcal{Q}_{\mathrm{GDL}}} \mathsf{Fld}/G, \qquad \mathsf{Fld} \xrightarrow{\mathcal{Q}_{\mathrm{CDL}}} \mathsf{Comp}.$$

In words: RSVP is the thermodynamic quotient of eventful implementations; GDL is the symmetry quotient of RSVP; CDL is the lawfulness quotient of RSVP. The ordering expresses the intended dependency: categorical and geometric architectural languages are extracted from an underlying field-theoretic description, itself extracted from implementation histories.

21 Conclusion: Constraints, Commitment, and the Boundary of Algebra

This paper has argued for a precise and limited claim: Categorical Deep Learning provides a complete and elegant theory of *lawful computation*, but lawfulness alone is insufficient to account for commitment, refusal, history, or agency. This is not a failure of the framework. It is the consequence of its success.

By unifying constraints and implementations through algebraic structure, categorical deep learning explains why certain architectures generalize, why weight tying and recursion are principled, and why invariances can be enforced without loss of expressivity. It shows that much of modern deep learning can be understood as the study of structure-preserving maps between parameterized computational spaces. In doing so, it replaces heuristic design with necessity.

At the same time, the analysis has shown that everything preserved by this unification is, by design, indifferent to history. Reparameterization symmetry, homomorphism requirements, and algebraic quotients all function by erasing implementation detail. Non-invertibility appears as information loss, not as obligation. Optimization produces convergence, not settlement. What survives is structure; what disappears is commitment.

The central contribution of this work is to make that boundary explicit and to locate it formally. Rather than attempting to enrich categorical deep learning with ad hoc notions of refusal or accountability, the paper reverses the usual direction of derivation. It begins from eventful implementations, shows how a field-theoretic description (RSVP) arises as a thermodynamic coarse-graining of history, and then demonstrates how both Geometric Deep Learning and Categorical Deep Learning emerge as distinct abstractions of that field-level description. Each abstraction forgets something different: geometry forgets energetic and entropic substance, while category theory forgets concrete event history.

Seen in this light, categorical deep learning is not an incomplete foundation, but a correct quotient. It captures exactly what remains invariant under lawfulness-preserving abstraction. What it cannot express—ownership, refusal, strong negation, and worldhood—are not bugs to be patched but structures that have already been erased by the act of abstraction itself.

The implications are twofold. First, expectations placed on purely algebraic accounts of intelligence should be recalibrated. If computation explains how systems act, it does not explain why they ever stop, give, or bind themselves to the consequences of action. Second, any attempt to build accountable or agent-like systems must either retain access to event history or reintroduce commitment as a first-class primitive. No amount of additional algebraic unification can recover what has been quotiented away.

In this sense, the title of the paper is literal. Constraints without commitment are not a deficiency of contemporary deep learning theory; they are its defining feature. Understanding that feature—where it comes from, what it explains, and what it necessarily excludes—is a prerequisite for any serious theory of agency beyond lawfulness.

A Algebraic Constraint Versus Commitment

Definition 13 (Algebraic Constraint)**.** *Let $\mathcal{C}$ be a category and $F : \mathcal{C} \to \mathcal{C}$ an endo-functor. An* algebraic constraint *is the requirement that a morphism $f : A \to B$ be a homomorphism between F-algebras, i.e., that the following diagram commute:*

$$\begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & B \end{array}$$

Such constraints specify lawful computation by preserving algebraic structure.

Definition 14 (Commitment)**.** *A* commitment *is an event that restricts a system's future admissible morphisms in a way that cannot be undone by reparameterization or equivalence. Formally, a commitment at time t induces a strict subcategory $\mathcal{C}_{t+1} \subset \mathcal{C}_t$ such that*

$$\mathrm{Hom}_{\mathcal{C}_{t+1}}(A, B) \subsetneq \mathrm{Hom}_{\mathcal{C}_t}(A, B)$$

for at least one pair (A, B).

Proposition 4. *Algebraic constraints do not imply commitment.*

Proof. Algebraic constraints restrict morphisms to those preserving structure but do not remove morphisms from the category itself. Reparameterizations via two-morphisms preserve the homomorphism condition and therefore leave the set of admissible morphisms invariant up to equivalence. No strict subcategory is induced. □

This establishes that algebraic lawfulness constrains computation without binding future action.

B Non-Invertible Computation and the Absence of Obligation

Definition 15 (Non-Invertible Morphism)**.** *A morphism $f : A \to B$ in a category $\mathcal{C}$ is* non-invertible *if there exists no morphism $g : B \to A$ such that $g \circ f = \mathrm{id}_A$.*

Definition 16 (Information Loss)**.** *A computation exhibits* information loss *if $|A| > |B|$ or if distinct elements of A are identified under f.*

Proposition 5. *Information loss does not constitute obligation.*

Proof. Non-invertibility characterizes mappings between representations, not relations between agents and futures. The destruction of information under f does not restrict the existence of alternative morphisms $f' : A \to B$ nor does it constrain future morphisms from B. Therefore no binding constraint on future action is imposed. □

A system may be logically irreversible while remaining temporally and normatively reversible.

This shows that CDL's move from groups to monoids suffices to model irreversible computation but not irreversible commitment.

C Parametric Maps and Symmetric Loss

Definition 17 (Category of Parametric Maps)**.** *The category* **Para** *has objects* (A, P)*, where* A *is a space and* P *a parameter space. Morphisms are functions*

$$f : P \times A \to B$$

and two-morphisms are reparameterizations $\phi : P \to Q$ *such that* $f = g \circ (\phi \times \mathrm{id}_A)$*.*

Definition 18 (Symmetric Loss)**.** *Loss is* symmetric *if parameter updates apply uniformly across* P *and no subspace* $P' \subset P$ *is rendered inaccessible by the update rule.*

Proposition 6. **Para** *admits only symmetric loss.*

Proof. All admissible updates in **Para** are reparameterizations preserving the homomorphism condition. There exists no mechanism to permanently exclude a parameter region while preserving the categorical structure. Thus no unilateral loss or ownership can be represented. □

Game-theoretic commitment devices cannot be expressed in **Para** without additional structure.

D Refusal and Strong Negation

Definition 19 (Weak Negation)**.** *Weak negation is the exclusion of actions due to optimization criteria, such as cost or probability, and may be reversed by changes in resources or context.*

Definition 20 (Strong Negation (Refusal))**.** *Strong negation is the permanent exclusion of an action from the admissible future action space, independent of optimization incentives.*

Proposition 7. *Categorical constraints encode weak negation but not strong negation.*

Proof. Categorical constraints restrict morphisms by lawfulness. If a morphism exists and preserves structure, it remains admissible regardless of cost or preference. Therefore no permanent exclusion based on refusal can be encoded. □

Refusal requires historical state not representable in purely compositional semantics.

E Events and Temporal Asymmetry

Definition 21 (Process)**.** *A process is a sequence of composable morphisms in a category.*

Definition 22 (Event)**.** *An event is a transformation that induces a strict reduction in the future admissible morphisms of the system.*

Proposition 8. *Categories model processes but not events.*

Proof. Categories are closed under composition. For any morphism $f : A \to B$, further composition is always defined when types match. No intrinsic notion of terminal exclusion exists. □

Worldhood requires structure beyond categorical composition.

A Formal Derivation of Geometric and Categorical Deep Learning from RSVP

This appendix provides a formal account of the derivation order claimed in the main text. In particular, it demonstrates that both Geometric Deep Learning (GDL) and Categorical Deep Learning (CDL) arise as abstractions of an intermediate field-theoretic description (RSVP), which itself arises as a quotient of eventful implementations. The derivation is expressed using categories, equivalence relations, and functors.

A.1 Eventful Implementations as a Free History Category

Let Ev be a set of primitive event types equipped with admissibility rules specifying which events may follow which.

Definition 23 (History Category)**.** *Define the category* Hist *as follows:*

- *Objects are implementation states.*
- *Morphisms $h : x \to y$ are finite admissible event sequences such that replaying h from x yields y.*
- *Composition is concatenation of event sequences.*
- *Identity morphisms are empty histories.*

Hist is a free category generated by event transitions. It records full execution lineage and is maximally concrete.

A.2 RSVP as a Thermodynamic Quotient

Let Fld be a category whose objects are RSVP field configurations $(\Phi, \mathbf{v}, S)$ and whose morphisms are admissible field evolutions.

Definition 24 (Thermodynamic Equivalence)**.** *Define an equivalence relation $\sim_{\mathrm{RSVP}}$ on objects and morphisms of* Hist *by*

$$x \sim_{\mathrm{RSVP}} y \quad \Longleftrightarrow \quad (\Phi_x, \mathbf{v}_x, S_x) = (\Phi_y, \mathbf{v}_y, S_y).$$

Proposition 9 (RSVP as a Quotient Category)**.** *There exists a functor*

$$\mathcal{Q}_{\mathrm{RSVP}} : \mathsf{Hist} \to \mathsf{Fld}$$

that factors through the quotient $\mathsf{Hist}/{\sim_{\mathrm{RSVP}}}$*.*

Proof. By construction, $\mathcal{Q}_{\mathrm{RSVP}}$ is constant on equivalence classes of $\sim_{\mathrm{RSVP}}$. Thus it descends uniquely to a functor on the quotient. □

RSVP therefore represents the maximal structure invariant under thermodynamic coarse-graining of event histories.

A.3 Geometric Deep Learning as Symmetry Quotient of RSVP

Let G be a group acting on Fld by field automorphisms.

Definition 25 (Symmetry Equivalence)**.** *Define $X \sim_G Y$ iff there exists $g \in G$ such that $Y = g \cdot X$.*

Definition 26 (Symmetry Quotient Category)**.** *Define Fld/G to be the orbit category whose objects are G-orbits $[X]$ and whose morphisms are induced by G-equivariant maps.*

Proposition 10 (GDL from RSVP)**.** *Equivariant neural architectures correspond precisely to morphisms in Fld/G.*

Proof. A map descends to the quotient iff it is invariant under the group action, i.e. equivariant. This is exactly the defining condition of layers in Geometric Deep Learning. □

Thus GDL is obtained by forgetting energetic and entropic detail while retaining symmetry invariants.

A.4 Categorical Deep Learning as Lawfulness Quotient of RSVP

Let Comp be a category of abstract computations (types and programs).

Definition 27 (Lawfulness Equivalence)**.** *Define $X \sim_{\text{law}} Y$ in Fld iff X and Y realize the same algebraic semantics, i.e. they satisfy the same pre/post-condition behavior under all admissible inputs.*

Proposition 11 (Existence of Lawfulness Quotient)**.** *There exists a functor*

$$\mathcal{Q}_{\text{CDL}} : \mathsf{Fld} \to \mathsf{Comp}$$

that identifies RSVP realizations up to $\sim_{\text{law}}$.

Proof. Algebraic semantics define congruence classes on realizations. Identifying realizations within each class yields the quotient category Comp. □

Within this quotient, architectures are expressed as algebras of endofunctors, and constraints are homomorphism conditions.

A.5 Incomparability of GDL and CDL

Proposition 12 (Orthogonality of Abstractions)**.** *The quotients Fld/G and Comp are generally incomparable: neither refines the other.*

Proof. Fld/G identifies realizations differing by symmetry actions but may distinguish realizations with identical computational semantics. Conversely, Comp identifies realizations with identical semantics regardless of symmetry. Therefore neither quotient factors through the other in general. □

A.6 Summary Diagram

The derivation order is summarized by the following diagram of quotient functors:

$$\mathsf{Hist} \xrightarrow{\mathcal{Q}_{\mathrm{RSVP}}} \mathsf{Fld} \xrightarrow{\mathcal{Q}_{\mathrm{GDL}}} \mathsf{Fld}/G, \qquad \mathsf{Fld} \xrightarrow{\mathcal{Q}_{\mathrm{CDL}}} \mathsf{Comp}.$$

Geometric Deep Learning and Categorical Deep Learning are thus extracted from RSVP by forgetting different structure. RSVP itself is extracted from implementation histories by forgetting event-level detail while retaining thermodynamic invariants.

B A Minimal Extension Admitting Commitment

This appendix defines a minimal categorical extension sufficient to represent commitment and refusal, without abandoning the compositional semantics of Categorical Deep Learning.

Definition 28 (Commitment-Enriched Category)**.** *Let $\mathcal{C}$ be a base category. A* commitment-enriched category $\mathcal{C}^\sharp$ *consists of:*

- *the objects and morphisms of $\mathcal{C}$;*
- *a distinguished class of* commitment morphisms $c : A \to A$*;*
- *a preorder $\preceq$ on morphisms such that $f \preceq g$ denotes that f is admissible whenever g is.*

Definition 29 (Commitment)**.** *A morphism $c : A \to A$ is a* commitment *if, for all $f : A \to B$,*

$$f \circ c \preceq f \quad \textit{but} \quad c \circ f \not\preceq f.$$

Intuitively, a commitment restricts future admissible actions but cannot be neutralized by precomposition.

Proposition 13. *$\mathcal{C}^\sharp$ is not equivalent to $\mathcal{C}$.*

Proof. In $\mathcal{C}$, admissibility is determined solely by type compatibility and lawfulness. In $\mathcal{C}^\sharp$, admissibility depends on historical commitments encoded by the preorder $\preceq$. Therefore no equivalence functor can preserve both structures. □

Commitment cannot be represented in purely compositional semantics.

This extension is minimal: no new objects are added, and composition remains defined, but admissibility becomes history-sensitive.

C Event-Recording Morphisms and Broken Reparameterization Symmetry

This appendix introduces a class of morphisms that record events explicitly and thereby break the reparameterization symmetry central to parametric map semantics.

Definition 30 (Event-Recording Morphism)**.** *Let* **Para** *be the 2-category of parametric maps. An* event-recording morphism *is a 1-morphism*

$$f : (A, P) \to (B, Q)$$

together with a map

$$\epsilon_f : P \to E$$

into an event space E, such that distinct parameterizations induce distinct event records.

Definition 31 (Event Sensitivity)**.** *A morphism f is* event-sensitive *if there exists no nontrivial reparameterization $\varphi : P \to P$ such that*

$$f \circ (\varphi \times \mathrm{id}_A) = f \quad \textit{and} \quad \epsilon_f \circ \varphi = \epsilon_f.$$

Proposition 14. *Event-sensitive morphisms break the reparameterization symmetry of* **Para***.*

Proof. In **Para**, two implementations differing only by reparameterization are identified by a 2-morphism. Event sensitivity ensures that such reparameterizations alter ϵ_f, preventing identification. Thus symmetry is broken. ☐

Weight tying and reparameterization cease to be purely structural when events are recorded.

Event-recording morphisms provide a formal mechanism for lineage, refusal, and irreversible commitment to appear within an otherwise algebraic framework.

D Relation to Categorical Deep Learning

This appendix situates the present work relative to the framework developed by Gavranović et al. in *Categorical Deep Learning is an Algebraic Theory of All Architectures*.

Proposition 15. *The present analysis is conservative with respect to Categorical Deep Learning.*

Proof. All constructions in the main text reduce to standard CDL semantics when commitment morphisms and event records are omitted. In particular, the base categories, endofunctor algebras, and parametric map semantics coincide exactly with those of Gavranović et al. ☐

Proposition 16. *Categorical Deep Learning is complete with respect to lawful computation.*

Proof. CDL characterizes architectures as algebras satisfying homomorphism conditions. These conditions fully determine which computations are lawful, invariant, and replayable under reparameterization. ☐

Proposition 17. *Categorical Deep Learning is incomplete with respect to commitment.*

Proof. CDL admits all reparameterizations preserving algebraic structure. Therefore no morphism can permanently restrict future admissible morphisms. Commitment and refusal, as defined in Appendix G, require admissibility relations not present in the CDL formalism. ☐

The present extensions are orthogonal to, not in competition with, CDL.

In summary, Categorical Deep Learning provides the correct algebraic account of architectural lawfulness. The present work identifies a minimal extension required to model historical binding, refusal, and accountability—phenomena intentionally outside the scope of CDL as originally formulated.

A Worked Example: Carry-as-Commitment and Event-Recorded Refusal

This appendix gives a concrete toy example showing (i) how an event-recording extension breaks reparameterization symmetry, and (ii) how a minimal commitment operator encodes refusal as strong negation. The example is intentionally small: a two-digit incrementer with carry. The point is not numerical performance but the categorical distinction between *lawful computation* and *settled history*.

A.1 Base computation: two-digit increment with carry

Let the state space be

$$A = \{0, \ldots, 9\} \times \{0, \ldots, 9\}, \qquad (a, b) \in A$$

representing a two-digit number $10a + b$.

Define the deterministic increment-with-carry function

$$\mathrm{inc} : A \to A$$

by

$$\mathrm{inc}(a, b) = \begin{cases} (a,\ b+1) & \text{if } b < 9, \\ (a+1,\ 0) & \text{if } b = 9 \text{ and } a < 9, \\ (0,\ 0) & \text{if } (a, b) = (9, 9). \end{cases}$$

The salient feature is the *carry event*: $b = 9$ forces a change in the higher digit.

In standard categorical or programming semantics, this is simply a function. It is lawful, total, and compositional.

A.2 A parametric implementation and its reparameterization symmetry

To illustrate the issue in parametric-map semantics, consider an implementation that depends on a parameter θ encoding a choice of representation for the digit pair. Let

$$P = \{0, 1\}$$

be a trivial "representation selector." For $\theta \in P$, define an implementation

$$f_\theta : A \to A$$

by

$$f_0 = \mathrm{inc}, \qquad f_1 = \sigma^{-1} \circ \mathrm{inc} \circ \sigma,$$

where $\sigma : A \to A$ is the digit-swap bijection $\sigma(a, b) = (b, a)$. Thus f_1 implements the *same computation* in a conjugate representation.

Package this in **Para** as the parameterized map

$$f : P \times A \to A, \qquad f(\theta, x) = f_\theta(x).$$

In ordinary **Para**, θ-choices related by a reparameterization are regarded as implementation variation. In particular, the nontrivial reparameterization $\varphi : P \to P$ given by $\varphi(0) = 1, \varphi(1) = 0$ witnesses that these are essentially the same family up to parameter relabeling.

This is precisely the symmetry we will now break.

A.3 Event recording: making carry an explicit trace

Let the event space be

$$E = \{0, 1\},$$

where 1 means "a carry occurred" and 0 means "no carry occurred."

Define the event recorder for the underlying increment:

$$\epsilon : A \to E, \qquad \epsilon(a, b) = \begin{cases} 1 & \text{if } b = 9, \\ 0 & \text{otherwise.} \end{cases}$$

Now define an event-recording morphism as an extension of f:

$$\widehat{f} : P \times A \to A \times E, \qquad \widehat{f}(\theta, x) = \big(f_\theta(x),\ \epsilon(x)\big).$$

Proposition 18 (Event recording breaks a reparameterization symmetry)**.** *The reparameterization $\varphi : P \to P$ does not identify $\widehat{f}$ with itself as a 2-morphism in the event-recording extension, unless it preserves the event record.*

Proof. In plain **Para**, relabeling θ can identify two implementations if their observable behavior agrees up to reparameterization. In the event-recording extension, equality requires agreement on both outputs: state and event.

But the event recorder depends on the *input* x (specifically on whether $b = 9$) and is therefore representation-sensitive. Under conjugation by σ, the carry predicate changes from "$b = 9$" to "$a = 9$" if one insists on recording carries in the original coordinate system. Consequently, the pair $(f_\theta(x), \epsilon(x))$ is not invariant under the parameter swap φ unless the event recorder is transformed in lockstep.

Thus the reparameterization that was admissible in **Para** no longer preserves the enriched observable, and the symmetry is broken. □

Interpretation: two implementations that are computationally equivalent can be distinguished by what they *record* as having happened. This is the minimal formal mechanism by which "biography" can appear.

A.4 Commitment as irreversible restriction of admissible continuations

We now add a commitment structure capturing strong negation. Let the enlarged state include a single commitment flag:

$$A^\sharp = A \times \{0,1\},$$

where the second component r records whether a refusal commitment is active.

Define a commitment morphism

$$c : A^\sharp \to A^\sharp$$

that activates refusal once a carry has occurred:

$$c\big((a,b),r\big) = \begin{cases} \big((a,b),1\big) & \text{if } b = 9, \\ \big((a,b),r\big) & \text{otherwise.} \end{cases}$$

Intuitively: "after you have carried, you are committed to a constraint."

Now define an admissibility rule for subsequent actions. Let

$$\mathsf{Step} = \{\text{inc}, \text{reset}\}$$

be two candidate next-actions, where $\text{reset}(a,b) = (0,0)$ is a "collapse" operation. Lift both actions to $A^\sharp$ by acting on the A-component and leaving r unchanged.

Define that reset is *forbidden* once refusal is active:

$$\text{reset admissible at } ((a,b),r) \quad \Longleftrightarrow \quad r = 0.$$

Equivalently, the commitment induces a strict reduction of future admissible morphisms.

Proposition 19 (Carry induces a commitment that forbids collapse)**.** *There exists an initial state $x_0 = ((a,9),0)$ such that after composing with c, the action* reset *is no longer admissible.*

Proof. Let $x_0 = ((a,9),0)$. Since $b = 9$, we have $c(x_0) = ((a,9),1)$. By definition of admissibility, reset is admissible iff $r = 0$, but here $r = 1$. Therefore reset is forbidden after the commitment. □

[Strong negation is history-sensitive] The forbiddenness of reset cannot be inferred from the instantaneous two-digit value (a,b) alone; it depends on the event-history summarized by the commitment flag.

Interpretation: a purely algebraic description of inc does not entail that reset is forbidden after carry. The prohibition arises only once events are recorded and commitments restrict admissible continuations.

A.5 What this example demonstrates

This toy system exhibits three separations that motivate the main text.

1. **Lawful computation vs. event biography.** The increment map is lawful and compositional; the event record makes “carry” an explicit trace element not quotiented away by reparameterization.

2. **Information loss vs. obligation.** The carry mechanism is a non-invertible feature of the transition, but obligation appears only after introducing commitment that restricts future admissible moves.

3. **Weak vs. strong negation.** A cost-based “back-off” could still allow reset if resources change. Here, reset is forbidden regardless of utility once refusal is active.

Thus, by adding (i) event-recording observables and (ii) a minimal commitment operator that reduces future admissible morphisms, one obtains a precise formal sense in which implementation histories can generate both the appearance of refusal and the breaking of reparameterization symmetry.

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